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“

Branding is what people say about you when you are not in the room.

”



-Jeff Bezos



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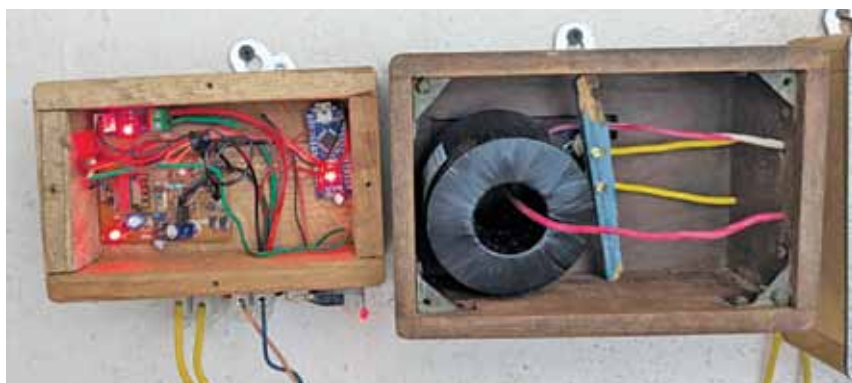


Fig. 5: Author's assembled circuit enclosed in a wooden box

value. The current is then gradually increased slightly above the preset full-load current (I_s), and the relay operation is observed after the programmed time delay.

Alternatively, the relay system may be connected to an actual load, as shown in the schematic. Under normal load conditions, the relay remains inactive. The load may be increased gradually using an external device such as an immersion heater or electric iron. The relay trips after the defined inverse-time delay characteristic.

Once the relay operates, it remains latched even if the load current falls to zero. Resetting is possible only by pressing the reset switch or by power-cycling the Arduino Nano.

This IDMT relay system can be used for alarm generation or load tripping, depending on the importance of the protected system.

In alarm mode, the relay output is used only to activate an audible or visual alarm when the current exceeds the preset limit, alerting the operator without disconnecting the load. This mode is suitable for

PARTS LIST	
<i>Semiconductors:</i>	
Board1	- Arduino Nano board
Sensor	- ACS712-30A current sensor
T1	- BC547 NPN transistor
D1, D2	- 1N4007 rectifier diode
LED1	- 5mm LED
<i>Resistors (all 1/4-watt, $\pm 5\%$ carbon):</i>	
R1	- 1k Ω
R2	- 10k Ω
R3	- 4.7k Ω
<i>Miscellaneous:</i>	
Jach1	- 12V DC socket
S1	- Push-to-on switch
CON1-CON4	- 2-pin connector
RL1	- 12V, SPDT relay
	- 12V, 1A adaptor
	- Jumper wire as required
	- Wooden enclosure
	- AC load
	- Current transformer, CT (optional)

critical systems where manual intervention is preferred.

In load-tripping mode, the relay output disconnects the load automatically during an overcurrent condition. Load tripping involves isolating the electrical load from the supply to prevent damage due to overheating or faults. For this purpose, the relay contacts are wired in series with a contactor or circuit breaker trip coil supplying the load. When the current exceeds the set value and the inverse-time delay expires, the relay energises, causing the contactor or breaker to open and safely disconnect the load from the supply. **EFY**



P. Balasubramanian is a retired Scientific Officer from Nuclear Power Corporation of India Ltd (NPCIL). His areas of interest include microcontrollers and power electronics.